



ANNEX 4

USV Phoenix

Technical Specification List

Particulars	
Vessel Type	Uncrewed Surface Vessel
Model	Phoenix Class
Classification	Diesel USV Patrol Vessel
Length	23ft / 7.27m
Breadth	2.95m
Draft	0.35m / 0.55m
Displacement	4,000kg (max)
Max Speed	100mph
Tank Capacity	400gallons / 1500litres

Powertrain / Propulsion	Price	Y/N
2 x COX V8 300HP Outboard Engines (Diesel) 250hour service interval	Included	Y
1 x Fisher Panda Marine AGT DC 48v Generator 8.0kW (extended service interval 200hours)	Included	Y
1 x Freshwater glycol system for generator seawater heat exchanger cooling	Included	Y
1 x 8 Jet Water Thruster Station Keeping System (100kgf) Dual Pump	Included	Y
Electrical		
2 x 48v Victron Lithium Battery System with BMS (2 x 280ah 24v) (Total 26.88kWh)	Included	Y
2 x 48v - 240vac Victron Quattro 8.0kW Inverter Chargers	Included	Y
1 x Electrical Consumer unit with failover switch for 240vac	Included	Y
1 x Victron Smart Cerbo GX System Manager with Display	Included	Y
Remote battery switches 48v, 24v and 12v	Included	Y
4 x DC - DC Step down converters 48v -24v (1.2kW)	Included	Y
2 x DC - DC Step down converters 48v -12v (360W)	Included	Y
2 x COX 24v Alternator 80A	Included	Y

2 x DC to DC Buck Boost 24v to 48v (100amps) for charging batteries from COX engine	Included	Y
2 x PLC Digital Input / Output Switching System (16 spare Inputs / Outputs) Redundant System (WAGO PLC)	Included	Y
Electrical looms and distribution systems	Included	Y
Galvanic Isolator for shore power charging	Included	Y
4 x 24v Bilge pumps (auto and manual activation)	Included	Y
2 x Bilge Low Level Sensors	Included	Y
1 x Fuel tank level sender	Included	Y
2 x Airconditioning systems (Dometic Turbo GTX 8000btu) 240v	Included	Y
2 x Ventilation system with shut off	Included	Y
2 x Internal Light	Included	Y
2 x NMEA2000 QNB System	Included	Y
Navigation		
1 x Solid State Digital Heading Compass (Maretron/Raymarine)	Included	Y
1 x GNSS primary system (survey grade) Hemisphere V123 (Heading +Position)	Included	Y
1 x GNSS secondary system (Simrad GS25) (Position)	Included	Y
2 x ProMux converter NMEA0183 to TCP (Position/Heading/Depth data)	Included	Y
1 x Radar System (X-Band) Raymarine Quantum 2	Included	Y
1 x Class A Marine AIS transponder (AIS4000 Raymarine) / SILENT MODE S/W	Included	Y
1 x Airmar Weather Station WX220	Included	Y
1 x Airmar USDST800 Ultrasonic Depth/Speed Log	Included	Y
1 x VHF radio (Icom) with remote access	Included	Y
1 x Deck Intercom/Loudspeaker/Horn System	Included	Y
Navigation Deck Lights (Port, Stbd, All round white) Lopo Lights LED (rated to 2nm)	Included	Y
RAM Lights (Red, White, Red) Lopo Lights LED (rated to 2nm)	Included	Y
Lopo Light LED detection system	Included	Y
Power / antenna loom fitted for Coastguard Tracking Device, but no device	Included	Y
Flashing Blue White Red Coastguard Light Bar	Included	Y
Vision		
4 x 130 degree external camera system	Included	Y
2 x Pay load cameras (AXIS)	Included	Y
1 x C014 Military Camera System	Added	Y
1 x Sea.AI Offshore Situational Awareness AI System	Included	Y
1 x Onboard NVR digital recorder 1TB hard rive	Included	Y
Communication / Networking		
1 x Silvus Stream caster 4400 Radio Link (400-2.7ghz) Range 20km+ 35Mbps Bandwidth	Included	Y
1 x THURAYA UAE satellite Marine system / VSAT THURAYA Equipment Only (Does not include subscriptions)	Included	Y
1 x 5G / LTE celular modem (Peplink MAX Br1)	Included	Y
1 x Mini server rack vertical mount	Included	Y

2 x Network Switch (Unifi USW-Pro-Max-24-PoE 400W)	Included	Y
1 x Network Gateway & Firewall +VPN (Unifi Dream Machine UDM-Pro-Max)	Included	Y
AutoPilot		
2 x .USV AutoPilot Computer System with dual redundancy	Included	Y
2 x Dometic Optimus Outboard Steering System (electric)	Included	Y
1 x Onboard HMI system	Included	Y
1 x Mission Planner Chart System with required Onboard Computer Server	Included	Y
Deck Equipment		
1 x Henriksen Lifting Hook HMKR3.5 Centre Lift	Included	Y
1 x Henriksen Lifting Hook HPHH2.5 Bow Claw	Included	Y
1 x Henriksen Lifting Ring HLR 3.5	Included	Y
2 x VHF Antenna	Included	Y
2 x Poynting Antenna (Wifi + Cellular)	Included	Y
1 x Eye tie down points (Bow)	Included	Y
2 x Eye tie down points (Stern)	Included	Y
4 x Deck cleats (3t)	Included	Y
2 x Pump out points	Included	Y
1 x Fuel fill point	Included	Y
3 x Deck flood lights	Included	Y
1 x Watertight entry hatch to vessel with closed/open sensor	Included	Y
1 x information panel with emergency stop switch and status	Included	Y
2 x Shore charging socket 32amps	Included	Y
2 x Mast deck gland (Roxtec)	Included	Y
1 x Ball valve deck vent for internal ventilation /exhaust	Included	Y
1 x Mooring lines set	Included	Y
Fire System		
Automatic Fire Suppression System (Stat-X or similar) (Tasneef MASS Code)	Included	Y
Fire Alarm System with at least two heat/smoke sensors	Included	Y
Lithium-Ion Battery Compartment / controlled Area (Tasneef Code)	Included	Y
4 x Compartment Temperature Sensors	Included	Y
Hull /Mast		
Composite twin hull catamaran built to Tasneef Standards and certification	Added	Y
Aluminium Mast Multi-Tiered	Included	Y
Painted in Camo Colours	Included	Y
Draft Marks	Included	Y
Vessel Identification Marks	Included	Y
USV Built to Tasneef standards and certification (MASS)	Added	Y
Documentation		

Technical Manual	Included	Y
Maintenance Manual	Included	Y
Stability Booklet ?????	Included	Y
General Arrangement Drawing	Included	Y
Mast Arrangement Drawing	Included	Y
Electrical Layout Drawing	Included	Y
Plumbing Layout Drawing	Included	Y
Network Drawing	Included	Y
FAT / SAT Test Document	Included	Y
Commissioning Report	Included	Y
Warranty		
7 Years Limited Warranty on hull, (water absorption)	Included	Y
12 months manufacturing warranty on / by 3 rd party vendors on supplied equipment	Included	Y
Remote Operation Centre		
1 x Control Desk setup for two Operator Stations (Setting up monitors /computers at your ROC, desks and chairs not provided)	Included	Y
1 x Large 85 inch Samsung QET Series (or similar) industrial signage screen for camera display	Included	Y
4 x Dell or similar Computer Monitors 19-22inch for console desk	Included	Y
4 x Dell Precision 7960 Workstation (or similar spec) with dual power supply / 1tb SSD / 64GB RAM / Nvidia RTX 4000 20GB		Y
4 x Keyboard and mouse Dell	Included	Y
2 x Tablet computers for dockside vessel control /marina operations	Included	Y
2 x Splashproof case for Tablet	Included	Y
Power connectors, power supplies, cabling, trunking equipment, wall mounting brackets	Included	Y
1 x Desktop Microphone and Speaker system	Included	Y
1 x Silvus Streamcaster 4400 Radio Link (400-2.7ghz) Range 20km+	Included	Y
1 x 5G / LTE celular modem (Peplink MAX Br1)	Included	Y
1 x Mini server rack vertical mount	Included	Y
2 x Network Switch (Unifi USW-Pro-Max-24-PoE 400W)	Included	Y
1 x Network Gateway & Firewall +VPN (Unifi Dream Machine UDM-Pro-Max)	Included	Y
1 x Video screen decoder system	Included	Y
Software for ROC		
4 x .USV Core Features Software (Lifetime license for unlimited control stations)	Included	Y
1 x Radio Operator Station for multiple VHF Channels +Intercom (Lifetime license – third party system for two PC's)	Included	Y
1 x Mission Planning Chart System (charts not included)	Included	Y
1 x Radar display software SDK (lifetime license)	Included	Y